# BASNet: Burned Area Segmentation Network for Real-Time Detection of Damage Maps in Remote Sensing Images

Weihao Bo, Jie Liu, Xijian Fan, *Member, IEEE*, Tardi Tjahjadi, *Senior Member, IEEE*, Qiaolin Ye, *Member, IEEE*, and Liyong Fu

***Abstract*— Since remote sensing images of post-fire vegetation are characterized by high resolution, multiple interferences, and high similarities between the background and the target area, it is difficult for existing methods to detect and segment the burned area in these images with sufficient speed and accuracy. In this article, we apply salient object detection (SOD) to burned area segmentation (BAS), the first time this has been done, and propose an efficient burned area segmentation network (BASNet) to improve the performance of unmanned aerial vehicle (UAV) high-resolution image segmentation. BASNet comprises positioning module and refinement module. The positioning module efficiently extracts high-level semantic features and general contextual information via global average pooling layer and convolutional block (CB) to determine the coarse location of the salient region. The refinement module adopts the CB attention module to effectively discriminate the spatial location of objects. In addition, to effectively combine edge information with spatial location information in the lower layer of the network and the high-level semantic information in the deeper layer, we design the residual fusion module to perform feature fusion by level to obtain the prediction results of the network. Extensive experiments on two UAV datasets collected from Chongli in China and Andong in South Korea, demonstrate that our proposed BASNet significantly outperforms the state-of-the-art SOD methods quantitatively and qualitatively. BASNet also achieves a promising prediction speed for processing high-resolution UAV images, thus providing wide-ranging applicability in post-disaster monitoring and management.**

***Index Terms*— Burned area segmentation (BAS), convolutional neural network (CNN), forest fire monitoring, salient object detection (SOD).**

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## I. Introduction

DURING the arid season of a year, about 400 Mha of vegetation-covered land in the world caught on fire [1]. As one of the natural phenomena, vegetation fires can promote vegetation regrowth, but they can also have a devastating impact on human infrastructure. In addition, secondary impact on the environment following fires can pose hazards such as erosion, flooding, and degradation of water quality [2], [3]. To minimize long-term hazards from vegetation fires, land management staff typically conduct post-disaster restoration soon after a fire and within a year [3]. Post-disaster management planning requires a structured assessment of the landscape mosaic and scale of the fire, which includes locating and estimating the extent of the burned area [4].

With the development of geo-information technology, satellite data can provide critical information for the management of vegetation fires. This information is spatially comprehensive and directly from surface features [5]. Currently, satellite data provided by Landsat, Modis and others have been applied in studies to assess post-fire vegetation cover resilience, vegetation recovery processes, etc. However, there are limitations in satellite data that cannot be ignored in such studies. For example, satellite image quality can be affected by cloud cover, and high-resolution images usually have a high cost [6]. Although new satellite sensors offer finer spatial scales and faster response capabilities to counter the drawbacks of satellite data being unsuitable for achieving regional and local targets, they are still expensive [7]. In contrast, unmanned aerial vehicles (UAVs) combine the advantages of high resolution, fast turnaround times, and lower operating costs with the ability to plan routes flexibly to avoid the effects of weather, thus providing wide-ranging applicability in post-disaster detection and management [8]. Today, remote sensing (RS) images acquired by UAVs are used in various image-processing tasks [9], [10]. Therefore, it also can be applied to the study of fire burned areas [11]. Fig. 1 shows examples from two datasets used in our study. Fig. 1(a) shows the damage map after a fire in Chongli, one of the three major competition areas of the 2022 Beijing Winter Olympics. Fig. 1(b) shows the damage map after a fire in South Korea.

There are many challenges for the effective retrieval of burned areas from UAV images. In addition, it is important to note that in the task of burned area segmentation (BAS),

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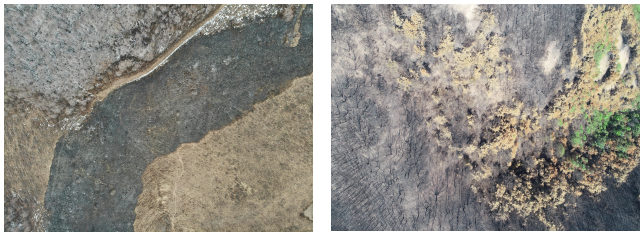


Fig. 1. Post-fire burned areas photographed by UAV. (a) China. (b) South Korea.

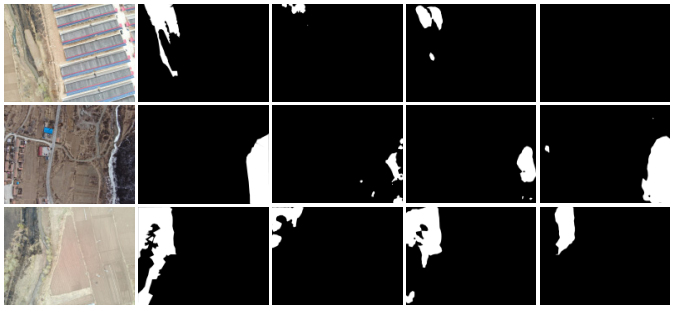


Fig. 2. Examples of the saliency maps of different methods. (a) Input images. (b) Ground truth. (c) UNet. (d) PSPNet. (e) DANet.

the main purpose is to detect the whole area that has been burned by fire using UAVs, which is of vital importance for post-disaster assessment and management. Although not belonging to the burned area, new growth and unburned soil are still within the scope of burned area, and thus it should be classified as target burned area that is useful for further assessment (The annotation of existing datasets also follows the same principle). For examples, Fig. 2 shows some examples of the saliency maps of different methods applied on the two datasets. The first challenge is the large scale of the burned area. As shown in Fig. 3(a), due to shooting time, weather or geography, there are many different interferences on a large area causing occlusions. Previous related studies have used semantic segmentation, which in fact is a pixel-level classification of the burned area, to address this challenge. For large regions with occlusions (e.g., new growth, unburned soil, etc.), and those affected by weather (e.g., frost, snow, etc.), semantic segmentation may classify these occlusions in the target region as other classes of pixels. Thus, the segmentation accuracy for the entire burned region is limited. To address this limitation, we employ salient object detection (SOD) for segmenting burned area. SOD is based on the human visual attention mechanism which aims to imitate this ability by automatically identifying and segmenting the most distinctive objects in an image [12]. Compared with semantic segmentation, SOD focuses more quickly on the more meaningful and salient information in the scene (i.e., the burned area in our case) while suppressing the influence of other irrelevant objects (e.g., the background including vegetation, houses, road, etc. and the occlusions within the burned area). Thus, it is advantageous for SOD to segment the burned area from UAV-acquired images with complex

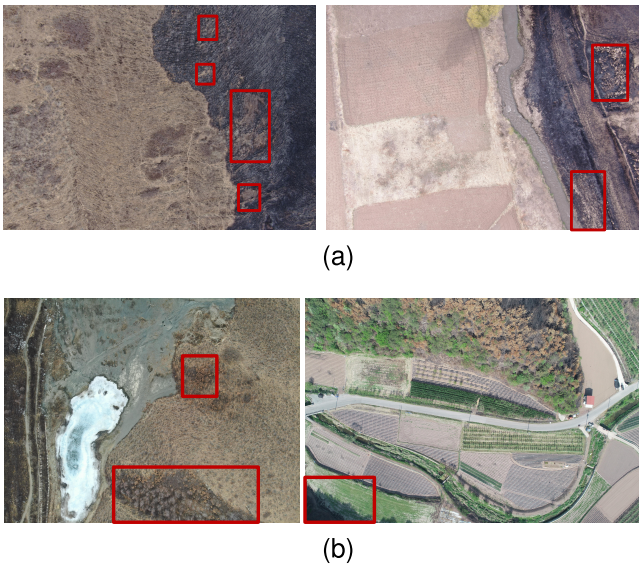


Fig. 3. Examples of the occlusions and similar patterns. (a) Examples of occlusions. (b) Examples of similar patterns.

background and occlusions [13], which helps to improve the robustness and accuracy of its segmentation. But these occlusions are also part of the burned area and need to be designated for post-disaster restoration. The SOD method we adopt segments the burned area as a complete target. To deal with scenes with more interfering objects, we propose to segment the significant targets in the image by combining global semantic and spatial information. There are two submodules in our network. The positioning module is used to identify the contextual information and rich semantic information of salient target regions in the high (i.e., deep) level of the encoder. The refinement module is used to identify spatial location and edge information in the low level, while the feature fusion is performed layer by layer using our residual fusion module (RFM) to finally segment the burned area as a whole target.

The second challenge is the complexity of region edges and numerous similar patterns of foreground and background in a large natural environment [as illustrated in Fig. 3(b)], posing difficulties for neural network learning. Therefore, we use the convolutional block attention module (CBAM) in the refinement module of the burned area segmentation network (BASNet). This enables BASNet to make fuller use of the edge texture and spatial location in the low and middle layers of the encoder. In addition, we employ an adaptive loss function. Usually binary cross-entropy (BCE) and intersection over union (IoU) are used globally for the loss functions. However, when all pixels are considered equally, these functions lead to class differences between foreground and background [14]. Therefore, we use the adaptive pixel intensity (API) loss, which in our case focuses more on pixels near the fine or explicit edges than on pixels in the background and in the center of the salient object. This ultimately allows our network to better segment the salient regions.

The third challenge is the high resolution of the images captured by UAVs. Traditional methods for BAS and machine

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learning require preprocessing and post-processing which fail to achieve end-to-end segmentation. The resolution of the images detected by general deep learning networks is also much lower than that of the images taken by UAVs. If these methods were applied to high-resolution images, then longer time and more equipment are required, rendering them less practical. Therefore, Tran *et al.* [11] used cropping to process such high-resolution images with some success. They cropped the images for prediction, and then combined the individual results to obtain the final prediction results. This resulted in two additional problems. First, the burned area caused by forest fires is generally large, and the segmented images may either include all burned areas or all background areas in a single image. Thus, the global spatial and semantic information could be lost when the deep learning network performs segmentation, hence reducing the accuracy of segmentation. Second, to identify large-scale regions in high-resolution images, inputting them to the deep learning network for prediction and followed by merging them require additional preprocessing and post-processing. Thus, the existing BAS methods cannot deal with RS images with high resolution effectively.

In this article, we address the challenges existing in the task of segmenting post-fire burned area using RS images. The major contributions of this article are summarized as follows.

1) An end-to-end network BASNet based on SOD is proposed to segment the burned area using high-resolution images acquired from UAV. BASNet effectively solves the problem that previous methods used for BAS cannot accurately detect the target in real time.

2) The positioning and refinement modules are proposed to capture the multilevel feature information by fusing rich semantics with the information of spatial location and edge, which effectively alleviate the problem of segmenting the large burned regions.

3) An adaptive loss function is introduced to enable the network to focus more on pixels near fine or explicit edges of burned regions and similar patterns between foreground and background in RS images, thus increasing the segmentation accuracy.

4) A high-resolution dataset containing UAV acquired image samples and the corresponding pixel-wise annotations is created from Chongli District for BAS. The proposed BASNet consistently outperforms other state-of-the-art methods in the experiments.

The rest of this article is organized as follows. In Section II, we briefly review the related works of SOD and its applications in optical RS images and BAS. In Section III, we present the details of the proposed BASNet. In Section IV, the experimental comparisons and ablation analysis are discussed. Finally, the conclusion is drawn in Section V.

## II. Related Work

### A. Salient Object Detection (SOD)

SOD generally refers to the detection and segmentation of salient objects/regions in a scene by imitating the attention mechanism of human vision [15]. Traditional SOD models rely on hand-crafted features or heuristic prior knowledge [12]. With the advent of deep learning, convolutional neural networks (CNNs) [16] with powerful feature-awareness have greatly contributed to the advancement of SOD. However, the performance of many SOD methods in practical applications is limited to resource conditions due to the size of the network with a large number of parameters and high-computational requirements. Shelhamer *et al.* [17] proposed a fully convolutional network (FCN), which substantially reduces computational cost while preserving spatial information by eliminating fully connected layers. FCN has inspired many SOD network architectures to be proposed. To integrate different layers of features, Hou *et al.* [18] introduced short connections of skiplayer structure in FCN. Wu *et al.* [19] proposed a cascaded partial decoder (CPD) framework that increases speed of operation by discarding low-level features and iteratively optimizing high-level features using saliency maps of high-level features. Recently, many networks are designed based on bottom-up architectures. These networks first acquire a coarse saliency map, then refine it by gradually combining spatial-detail-rich features at the lower layers, and finally output the final saliency map at the top layer [12]. Zhou *et al.* [20] created complementary trilateral decoder network (CTDNet) comprising three branches with three fusion modules from coarse to fine to improve the localization of a region and the quality of its boundary. They proposed a boundary refinement module (BRM) to utilize the salient boundary information provided by the boundary path to further refine the boundary. As an efficient and scalable preprocessing technique, SOD is nowadays applied to many downstream vision tasks, such as scene classification, visual tracking, light field image segmentation, RS image processing, etc.

SOD has been widely used not only in images of natural scenes, but also increasingly in optical RS images. Zhao *et al.* [21] proposed a sparsity-guided RS image SOD model for optical RS images. Zhang *et al.* [22] proposed an adaptive multiple feature fusion model using multiple feature relationships to improve the accuracy in the extraction of the region of interest (ROI) of SOD models for optical RS images. Li *et al.* [13] proposed an end-to-end deep network, LVNet, based on the shape of the network structure. LVNet detects significant objects in optical RS images in a purely data-driven manner. Zhang *et al.* [23] proposed a dense attentional fluid network to make shallow attentional cues flow into deeper layers to guide the generation of advanced feature attention maps such that they can better detect small targets in optical RS images. They employed CBAM [24] in the network structure to generate a single-scale attention map, which enhances the prediction effect.

However, current research on SOD for optical RS images tends to focus on small objects, and there has not been any relevant research on using SOD for large area detection. Compared with detecting small targets in RS images, SOD in large areas faces difficulties in the presence of occlusions in large areas, similar foreground and background information, and complex decomposition of edges. Thus, previous SOD

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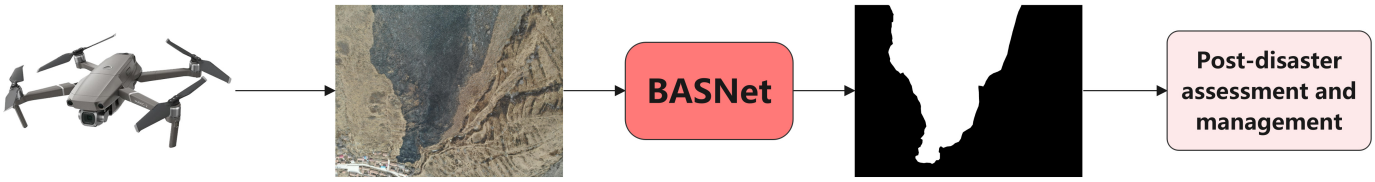


Fig. 4. Process for real-time BAS using an UAV to photograph the post-fire scene.

networks usually show unsatisfactory performance when dealing with large salient area detection.

### B. Burned Area Segmentation (BAS)

Several studies have proposed solutions for post-forest fire monitoring. Biase and Laneve [25] used an improved Satellite FIre DEtection (SFIDE) algorithm for BAS in satellite images. Brand and Manandhar [26] used an improved version of UNet to train and predict using the burned area dataset captured by Sentinel-2. Zanetti *et al.* [27] proposed a detection system based on the burn severity index threshold applied to multispectral images. However, the use of such satellite images can be affected by the correction period of satellite platforms, cloud coverage effects, projection shadows [6], and data acquisition. Therefore, several methods [11], [28], [29], [30], [31] used UAV images instead. The integration of UAV image segmentation algorithms has become an attractive research topic in the field of RS. It can be an effective tool for monitoring the damaged areas after forest fires. However, current methods for segmenting UAV images tend to reduce the size of their input images by resizing or cropping before training due to the limitation of equipment, where the semantic or spatial information might be lost, and extra post-processing is needed. Tran *et al.* proposed a dual segmentation network based on UNet [11]. They cropped the $5472 \times 3648$ images into $912 \times 912$ and $128 \times 128$, and then sent them to the network, requiring post-processing of the predicted results into red green blue (RGB) and EXIF data necessary for the DroneDeploy platform. In this work, the proposed network can directly process the RS images with high resolution without significant reduction of the size of the original images, avoiding the loss of useful information.

## III. Proposed Method

Fig. 4 shows the process of the proposed method. The map of the burned area (i.e., the damage map) is obtained in real time using a UAV to photograph the post-disaster area and simultaneously transfer it to the deployed BASNet. An appropriate staff can instantly determine the next priority flight area of the UAV based on the map, making the data collection and processing more flexible and rapidly. This process is not possible with existing methods used for BAS. The rest of this section is organized as follows. Section III-A describes the overall structure of BASNet. Section III-B and III-C introduce the two submodules of BASNet. Section III-D describes the loss function used by BASNet and the calculation of the total loss of the network output.

### A. Main Framework of BASNet

In this work, we propose a lightweight network BASNet specifically for BAS as shown in Fig. 5. For the encoder, we use ResNet-18 [32] as a backbone network. It is pretrained on ImageNet [33] and forms several feature layers. Due to the high resolution, the coarse lower level features require more computation, so we only utilize features from the last four stages with strides of 4, 8, 16, and 32 relative to the input image. These four features are denoted as $F^{(1)}$, $F^{(2)}$, $F^{(3)}$, and $F^{(4)}$.

The decoder of the network has two submodules: positioning module and refinement module. The positioning module captures rich semantic contextual information and global contextual information with large receptive fields from $F^{(3)}$ and $F^{(4)}$, and generates an initial coarse saliency mapping. The refinement module uses the CBAM to capture the spatial location and edge information in $F^{(1)}$ and $F^{(2)}$ to further refine the target and generate the final salient region map. The positioning module and the refinement module complement each other by fusing layer by layer via the RFM, obtaining the final output via the BRM. An auxiliary predictor exists at the end of the positioning module and the refinement module, enabling the network to pay more attention to pixels that are more difficult to predict, e.g., near fine edges or explicit edges via API loss. As a result, the localization accuracy of the region and the quality of the boundary are significantly improved.

We deploy a mask prediction layer for the decoded features and use a sigmoid layer to map the range of saliency scores into [0,1]. The prediction layer converts 3-D features into 2-D saliency maps based on the number of feature channels at different decoding scales. The final output of BASNet is from the predicted saliency map at the highest decoding level. The overall processing of the input image by BASNet is shown in Algorithm 1, where further details are given in Section III-B and III-C.

### B. Positioning Module

The currently available deep learning-based methods for BAS [11] tend to adopt the U-shape structure proposed by UNet [34]. However, the U-shape structure suffers from two problems [20]: 1) Dilution of semantic information in the process of layer feature fusion and 2) Weak capacity for capturing global contextual information. These problems may lead to inaccurate locations of salient objects and incomplete segmentation results. But semantic information and global contextual information are important for BAS due to the influence of: 1) occlusions in certain regions and 2) the similarity between background and foreground in the natural environment.

To address the two problems of U-shape structure, we employ the positioning module to capture rich semantic and global contextual information with large receptive fields,

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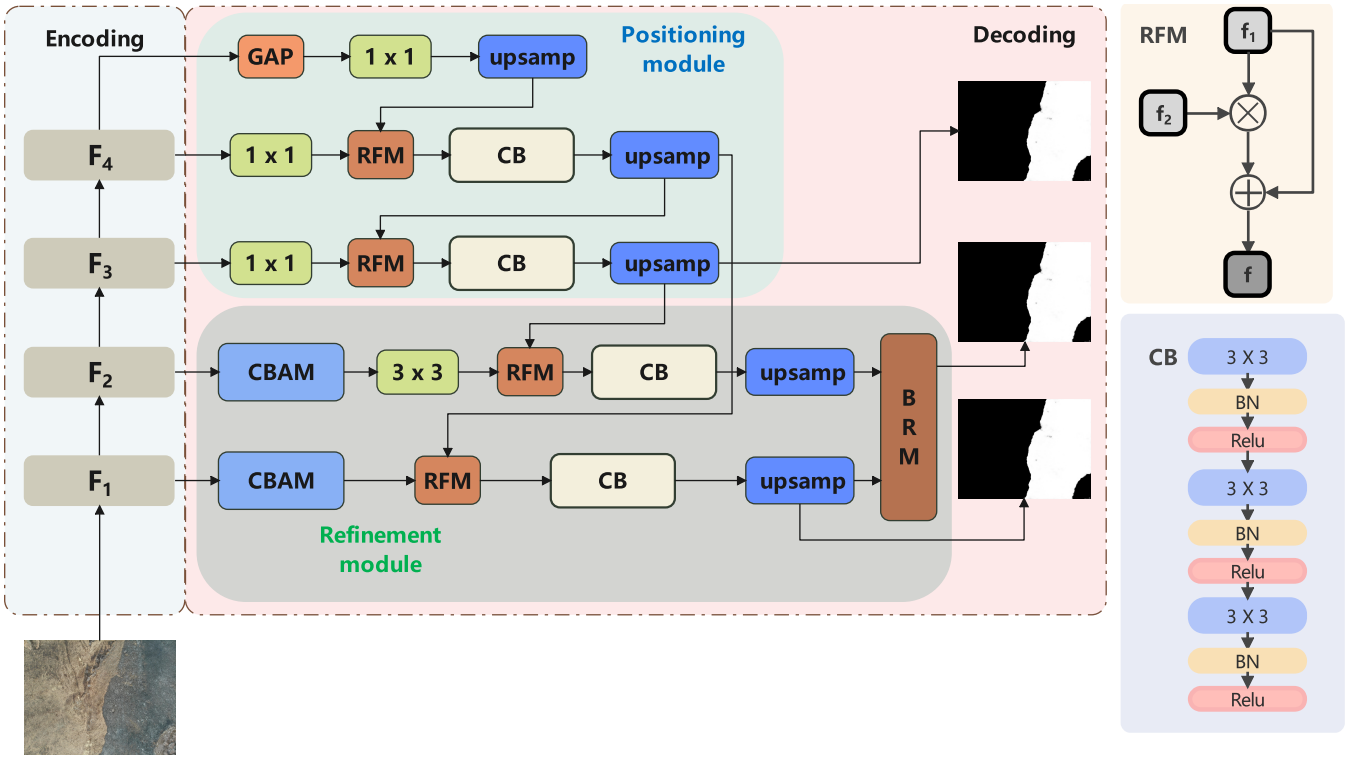


Fig. 5. Framework of the proposed BASNet which comprises a positioning module and a refinement module to, respectively, generate the initial coarse salient map and the final saliency region map.

**Algorithm 1** Pseudocode of BASNet

**Input:** $n$ burned area datasets $[S_1, \ldots, S_n]$

**Output:** Saliency maps $[F_1, \ldots, F_n]$, where $F_i$ includes $\left[P_i^{(1)}, P_i^{(2)}, P_i^{(3)}\right]$

Pretrain the encoder of the network and apply it on a test image

The four predicted feature layers are fed into the network decoder to generate the significance map

**for** $i = 1; i \leq n; i \leftarrow i + 1$ **do**

$F_i^{(1)} + F_i^{(2)} + F_i^{(3)} + F_i^{(4)} \leftarrow \mathrm{En}(S_i)$

Input feature layers $F_i^{(3)}$, $F_i^{(4)}$ into the positioning module (PM) to generate the predicted saliency map $P_i^{(1)}$:

$P_i^{(1)} \leftarrow \mathrm{PM}\left(F_i^{(3)}, F_i^{(4)}\right)$

Input feature layers $F_i^{(1)}$, $F_i^{(2)}$ into the refinement module (RM) to generate the predicted saliency map $P_i^{(2)}$:

$P_i^{(2)} \leftarrow \mathrm{RM}\left(F_i^{(1)}, F_i^{(2)}\right)$

Input $P_i^{(1)}$, $P_i^{(2)}$ into the BRM after feature fusion processing to generate the final predicted saliency map $P_i^{(3)}$:

$P_i^{(3)} \leftarrow \mathrm{BRM}\left(P_i^{(1)}, P_i^{(2)}\right)$

$F_i \leftarrow \left[P_i^{(1)}, P_i^{(2)}, P_i^{(3)}\right]$

**end for**

**return** $[F_1, \ldots, F_n]$

while integrating them via the RFM. The positioning module generates an initial mapping for the location of the salient regions as follows. First, to obtain more global contextual information in the positioning module, we use a global average pooling (GAP) layer at the tail of the backbone network. We then apply $1 \times 1$ convolution to $F^{(3)}$ and $F^{(4)}$ to limit the number of channels to 64, and remove the redundant information generated by too many channels to reduce the computational cost. The application of $1 \times 1$ convolution also facilitates the reduction of redundant information generated by subsequent modules of our network such as convolutional block (CB) and CBAM in processing images.

Finally, we propose an RFM and embed it into BASNet as shown in Fig. 5 that effectively fuses the upsampled output of the GAP layer and features of the latter two stages to form an initial coarse saliency mapping of the significant burned area. The positioning module is formalized as

$$M^{(1)} = \mathrm{CB}_1(\mathrm{RFM}_1(C_{1\times1}(F^{(4)}), Up_1(C_{1\times1}(\mathrm{GAP}(F^{(4)}))))) \tag{1}$$

$$M^{(2)} = \mathrm{CB}_2(\mathrm{RFM}_2(C_{1\times1}(F^{(3)}), Up_1(M^{(1)})) \tag{2}$$

$$P^{(1)} = Up_2(M^{(2)}) \tag{3}$$

where $C_{1\times1}$ and $Up$, respectively, denote the $1 \times 1$ convolution and bilinear interpolation upsampling. $Up_1$ denotes up-sampling to the size needed for the next RFM, and $Up_2$ denotes up-sampling to the same size as the original image for regression with masks. $P^{(1)}$ denotes the output of positioning module.

RFM is a residual fusion module. It fuses the result $f_2$ of the previous feature map processed by the decoder with the next feature map $f_1$ in the form of residual concatenation. It facilitates the exchange of iterative information between branches and mitigates the differences between features while enhancing

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the low-level features. We designed a set of weighting factor $\delta$ for feature fusion to balance the contribution of two parts. The parameters of 0, 0.2, 0.5, 0.7, 1, and 1.2 were selected for coefficients $\delta$. Based on the observation, we found that 1 is the optimal value in our BAS scenario. This allows RFM to reduce the redundant information introduced between each other when performing feature fusion, where $f_1$ and $f_2$ absorb useful information to complement each other. At the same time, redundant noise information is suppressed so that the entire model branch gradually generates more accurate salient feature maps based on the burned area as a whole when feature fusion is continuously performed. The RFM is formalized as

$$\mathrm{RFM}(f_1, f_2) = f_1 + \delta \cdot (f_1 \odot f_2) \tag{4}$$

where $\odot$ denotes the element-level multiplication between matrices, $\delta$ is the weighting factor, $f_1$ is the feature input of the current branch, and $f_2$ is the feature input of the previous branch.

To further integrate semantic information from the feature maps fused by RFM, we designed CB. In our design, each CB unit contains three convolutional layers, where the number of input and output channels is kept the same. Each convolutional layer is followed by a batch normalization layer and a ReLU activation function.

### C. Refinement Module

The positioning module determines the basic location of salient regions by capturing rich semantic contextual information and high-level feature information. However, since the high-level feature layer has been down-sampled several times, it loses much spatial location information and global information. Thus, it is difficult for the positioning module to obtain accurate edges and it might identify some false positive areas similar to the burned areas. Additionally, in previous deep learning-based segmentation of burned areas, researchers often use semantic segmentation networks based on U-shape. But often a U-shape type network has a large number of parameters and is difficult to be trained and to predict directly on a device like a 1080 Ti graphics processing unit (GPU). Thus, researchers would first subdivide the images and stitch the processed subimages to get the prediction results. We find that the segmentation maps produced in this way often have very rough boundaries. As shown in Fig. 6, the left figure [11] shows the latest research results on BAS in December 2020, where a network based on two-stage UNet semantic segmentation performed BAS. It can be seen that the prediction of the edges is relatively coarse. Therefore, we use the refinement module to extract the rich spatial location information and edge information in the lower layers of the network to obtain the results of BASNet on the right of Fig. 6, where the completeness of the segmentation and the accuracy of the edges are greatly improved.

In the refinement module, a CBAM attention module [24] is embedded behind $F^{(1)}$, $F^{(2)}$ of the backbone network. As shown in Fig. 7, the left panel shows the results without the addition of CBAM prediction, and the right panel shows the results after the addition. It can be seen that CBAM reduces the identification of false positive regions.

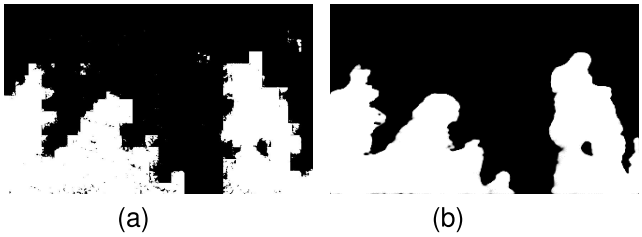


Fig. 6. Comparison of UNet and BASNet segmentation effect. (a) UNet. (b) BASNet.

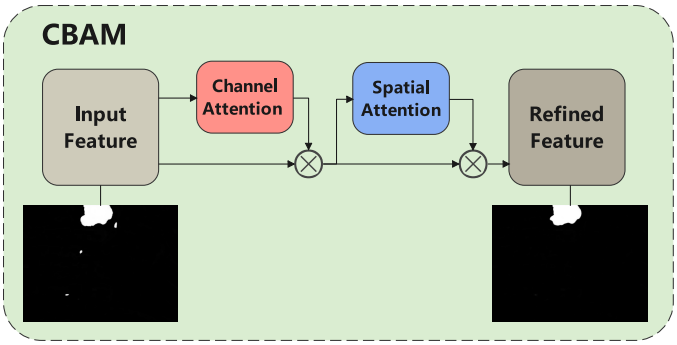


Fig. 7. CBAM.

The CBAM module will infer the attention maps sequentially along two independent dimensions (channel and spatial). This is followed by multiplying them with the input feature maps to perform adaptive feature optimization. The branch where $F^{(2)}$ is located then uses a $3 \times 3$ convolution after the CBAM module to reduce the number of channels and remove redundant information. Using a $3 \times 3$ convolution kernel instead of $1 \times 1$ better utilizes the spatial location information obtained from the CBAM module. The number of channels in the branch where $F^{(1)}$ is located is already the same as the other branches, so there is no need to reduce its dimension. $F^{(1)}$ and $F^{(2)}$ are then fed into RFM for fusion with the feature information obtained from the high level. It should be particularly noted that the feature layer fused with $F^{(1)}$ originates from $F^{(4)}$. And the semantic information of the high level is used to suppress the nonsignificant background region information of the low level to obtain the prediction map $P^{(2)}$, which is the bottom of the three outputs in Fig. 5 and $M^{(3)}$ which is from $F^{(2)}$. Finally, $M^{(3)}$ is fed into the BRM with $M^{(4)}$ for a more refined feature fusion to obtain the final prediction result $P$ which is the middle of the three outputs. The refinement module is formalized as

$$M^{(3)} = \mathrm{CB}_3(\mathrm{RFM}_3(C_{3\times3}(\mathrm{CBAM}_1(F^{(2)})), Up_1(P^{(1)}))) \tag{5}$$

$$M^{(4)} = \mathrm{CB}_4(\mathrm{RFM}_4(\mathrm{CBAM}_2(F^{(1)}), Up_1(M^{(1)}))) \tag{6}$$

$$P^{(2)} = Up_2(M^{(4)}) \tag{7}$$

$$P = Up_2(\mathrm{BRM}(Up_1(M^{(3)}), Up_1(M^{(4)}))) \tag{8}$$

where CBAM denotes CBAM attention module, RFM denotes residual feature module, BRM is the boundary refinement module, $P^{(2)}$ is the auxiliary predictor of the refinement module, and $P$ is the final prediction result of the network.

BRM is the module proposed by Zhao *et al.* [20]. It acts as the final fine fusion at the end of BASNet and refines the

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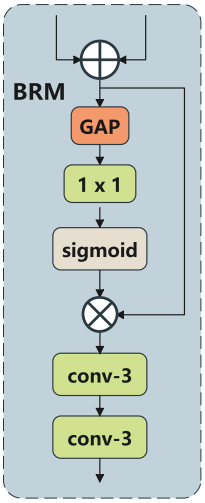


Fig. 8. Diagram of BRM.

predictions of the salient regions. As shown in Fig. 8, the output features of these two branches are first combined by an addition operation. The fused feature set $B_f$ is then generated as a feature vector. The attention vector is further computed, and feature learning is guided by $1 \times 1$ convolution and sigmoid functions. The weight vector can reweight the fused features $B_f$ for feature selection, and refinement by multiplication operations to generate $B_r$. Finally, the improved features $B_r$ are combined with $B_f$ to enhance the feature representation by two $3 \times 3$ convolutional layers. It is noted that each $3 \times 3$ convolution is followed by a batch normalization and a ReLU activation function. The above process is formalized as

$$B_f = Up_3(f_1) + f_2 \tag{9}$$

$$B_r = B_f \otimes \sigma\left(C_{1\times1}\left(\mathrm{GAP}\left(B_f\right)\right)\right) \tag{10}$$

$$\mathrm{BRM}(f_1, f_2) = C_{3\times3}\left(C_{3\times3}\left(B_r + B_f\right)\right) \tag{11}$$

where $Up_3$ denotes up-sampling to the same size as $f_2$, $\otimes$ denotes multiplication operation, and $\sigma$ denotes the weight vector.

### D. API Loss

For the loss function, we combined BCE loss and IoU loss to initially reduce the difference between the object and the background. However, BCE and IOU consider each pixel as globally equal, which can lead to class differences between foreground and background. The localization accuracy of an edge has been a difficult problem in BAS. Pixels close to the edge of the target region need more attention. Therefore, we used API loss [14], which applies pixel intensity $\omega$ to each pixel as

$$\omega_{ij} = (1-\lambda)\sum_{k\in K}\left|\frac{\sum_{h,w\in A_{ij}} y^k_{hw}}{\sum_{h,w\in A_{ij}} 1} - y_{ij}\right| y_{ij}. \tag{12}$$

The pixel intensity $\omega$ is used for adaptive BCE loss (aBCE) as (13), as shown at the bottom of the next page, where $y$ and $\hat{y}$, respectively, denote the label and prediction probability corresponding to the binary class. Compared with BCE loss, using $\omega$, aBCE loss enables the network to focus more on local structures associated with explicit or fine edges. In addition, the adaptive IoU (aIoU) loss optimizes the global structures based on the intensive features corresponding to $\omega$. The pixels which are highly associated with the intensive regions are discriminated and emphasized compared to the original IoU loss, that is

$$\mathcal{L}^a_{\mathrm{IoU}} = 1 - \frac{\sum_i^H \sum_j^W (y_{ij}\hat{y}_{ij})(1+\omega_{ij})}{\sum_i^H \sum_j^W (y_{ij} + \hat{y}_{ij} - y_{ij}\hat{y}_{ij})(1+\omega_{ij})}. \tag{14}$$

To incorporate the above local and global structural strengths, we combined the API loss functions as

$$\mathcal{L}_{\mathrm{API}}(y, \hat{y}) = \mathcal{L}^a_{\mathrm{BCE}}(y, \hat{y}) + \mathcal{L}^a_{\mathrm{IoU}}(y, \hat{y}). \tag{15}$$

The total loss function is denoted as

$$\mathcal{L}(P, G) = \mathcal{L}_{\mathrm{API}}(P^{(1)}, G) + \mathcal{L}_{\mathrm{API}}(P^{(2)}, G) + \mathcal{L}_{\mathrm{API}}(P, G). \tag{16}$$

## IV. Experiments

### A. Datasets and Evaluate Metrics

The proposed BASNet is evaluated on two datasets captured by UAVs: Chongli District dataset, China; and the Andong City dataset, South Korea. Chongli District dataset was created in February and May 2021 consisting of 104 RGB images with resolution $5174 \times 3888$ and $4056 \times 3034$ pixels. The Andong City dataset was created in May 2020 two weeks after a forest fire and is composed of 85 RGB images of resolution $5472 \times 3648$. Each dataset is randomly divided into training set and test set in the ratio of 7:3.

Three metrics are chosen to evaluate the performance of BASNet and other state-of-the-art networks. The first is mean absolute error (MAE), which represents the pixel-wise average absolute difference between the predicted and ground truth, that is

$$\mathrm{MAE} = \frac{1}{W \times H}\sum_{i=1}^{H}\sum_{j=1}^{W}|P(i,j) - G(i,j)| \tag{17}$$

where $W$ and $H$ are, respectively, the width and height of the image, while $P$ and $G$, respectively, denote the prediction mask and ground truth. The F-measure ($\mathrm{F}_\beta$) takes both precision and recall into account, that is

$$\mathrm{F}_\beta = \frac{(1+\beta^2)\ \mathrm{Precision}\ \times\ \mathrm{Recall}}{\beta^2 \times\ \mathrm{Precison}\ + \mathrm{Recall}} \tag{18}$$

where $\beta^2$ is set to 0.3, as is recommended in [35]. In addition to quantifying the pixel-level error between the predicted and ground truth, we also used E-measure ($\mathrm{E_m}$) [36] to measure the structural similarity between the two, that is

$$\mathrm{E}_m = \frac{1}{W \times H}\sum_{i=1}^{W}\sum_{j=1}^{H}\phi_P(i,j) \tag{19}$$

where $\phi_P$ is the enhanced alignment matrix.

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TABLE I
QUANTITATIVE COMPARISONS WITH STATE-OF-THE-ART SOD MODELS ON OUR TWO DATASETS, WHERE VALUES IN BOLD DENOTE BEST PERFORMANCES

| Methods | | CHINA | | | SOUTH KOREA | | | Resize |
|---|---|---|---|---|---|---|---|---|
| | | $F_\beta$ | MAE | $E_m$ | $F_\beta$ | MAE | $E_m$ | |
| Semantic segmentation | UNet | 0.491 | 0.097 | 0.494 | 0.283 | 0.190 | 0.269 | $896 \times 896$ |
| | PSPNet | 0.731 | 0.021 | 0.729 | 0.513 | 0.019 | 0.544 | $1024 \times 1024$ |
| | DANet | 0.608 | 0.062 | 0.617 | 0.566 | 0.021 | 0.643 | $640 \times 640$ |
| SOD | F3Net | 0.596 | 0.093 | 0.601 | 0.427 | 0.058 | 0.513 | $3072 \times 7072$ |
| | LDF | 0.632 | 0.070 | 0.642 | 0.477 | 0.043 | 0.549 | $3072 \times 7072$ |
| | CTDNet | 0.755 | 0.020 | 0.749 | 0.596 | 0.023 | 0.642 | $3072 \times 7072$ |
| | MINet | 0.594 | 0.069 | 0.618 | 0.475 | 0.038 | 0.573 | $2048 \times 2048$ |
| | GateNet | 0.673 | 0.036 | 0.715 | 0.466 | 0.045 | 0.543 | $2048 \times 2048$ |
| | CPD | 0.642 | 0.102 | 0.632 | 0.485 | 0.112 | 0.526 | $3072 \times 7072$ |
| | GLSTR | 0.671 | 0.038 | 0.705 | 0.395 | 0.065 | 0.471 | $384 \times 384$ |
| | GCPANet | 0.732 | 0.023 | 0.760 | 0.463 | 0.052 | 0.537 | $1024 \times 1024$ |
| | BASNet(ours) | **0.772** | **0.010** | **0.767** | **0.615** | **0.016** | **0.657** | $3072 \times 7072$ |

### B. Implementation Details

We use Pytorch to implement the proposed method. The experiments are performed on an NVIDIA 1080Ti GPU. For training, the images are resized to $3072 \times 3072$ pixel, and random cropping and random horizontal flipping are used for data augmentation. Due to memory limitations, some of the images are reduced in size during training, and the specific resize values are shown in Table I. During prediction, the images are also resized to $3072 \times 3072$. The prediction results are the desired saliency maps, and no post-processing is involved. Due to the GPU memory limitation, we adopt ResNet-18, pretrained on ImageNet, as the backbone network. The optimization method is Adam with a learning rate of 1e-3, weight decay of 1e-4, batch size of 3, and training for 60 epochs.

It should be noted that considering the actual deployment scenario of BAS, we use the video memory (12 G) to more than 90% of the maximum computing capacity of the 1080Ti during the training process. However, current state-of-the-art SOD networks are mostly designed for small images and therefore require a very large amount of GPU memory to predict high-resolution images. To make the training and prediction process meet the needs of general equipment, there are generally two ways of processing: 1) cropping large images into multiple small images and feeding them into the network for prediction and 2) resizing large images to smaller ones for prediction. We conducted experiments on BASNet in both ways. As shown in Table II, resizing the image is better than cropping the image for the experiment. In addition, the smaller the cropped images, the worse the network performs. The reason is that since the burned area is very large, there will be a large number of cropped images that are either all target or nontarget regions. A large amount of global contextual information and spatial location information are lost, resulting in very poor prediction results. If resizing is used, the closer a resized image is to its original image, the better the network performs. Therefore, we adjusted the resize value in the subsequent comparison experiments to cope with the GPU memory demand of the different networks, so that it fits onto the 12 G memory in NVIDIA 1080Ti.

TABLE II
QUANTITATIVE COMPARISON AFTER RESIZING OR CROPPING ON THE TWO DATASETS

| Operations | Size | CHINA | | | SOUTH KOREA | | |
|---|---|---|---|---|---|---|---|
| | | $F_\beta$ | MAE | $E_m$ | $F_\beta$ | MAE | $E_m$ |
| Resizing | $2048 \times 2048$ | 0.758 | 0.021 | 0.777 | 0.565 | 0.024 | 0.62 |
| | $1024 \times 1024$ | 0.729 | 0.022 | 0.756 | 0.522 | 0.025 | 0.597 |
| | $512 \times 512$ | 0.688 | 0.025 | 0.741 | 0.547 | 0.028 | 0.633 |
| | $3072 \times 3072$ | **0.772** | **0.010** | **0.767** | **0.615** | **0.016** | **0.657** |
| Cropping | $2048 \times 2048$ | 0.472 | 0.019 | 0.431 | 0.384 | 0.026 | 0.311 |
| | $1024 \times 1024$ | 0.304 | 0.017 | 0.206 | 0.237 | 0.019 | 0.187 |
| | $512 \times 512$ | 0.097 | 0.019 | 0.102 | 0.125 | 0.023 | 0.119 |
| | Original | **0.772** | **0.010** | **0.767** | **0.615** | **0.016** | **0.657** |

### C. Comparison With the State-of-the-Art SOD Methods

To verify the effectiveness of the proposed method, we compared it with eight state-of-the-art SOD methods: CTDNet [20], F3Net [37], label decoupling framework (LDF) [38], multi-scale interactive network (MINet) [39], global context-aware progressive aggregation network (GCPANet) [40], GateNet [41], CPD [19], and Global-Local

$$\mathcal{L}^a_{\text{BCE}} - \frac{\sum_i^H \sum_j^W (1+\omega_{ij}) \sum_{c=0}^1 (y_c \log(\hat{y}_c) + (1-y_c)\log(1-\hat{y}_c))}{\sum_i^H \sum_j^W (1.5+\omega_{ij})} \tag{13}$$

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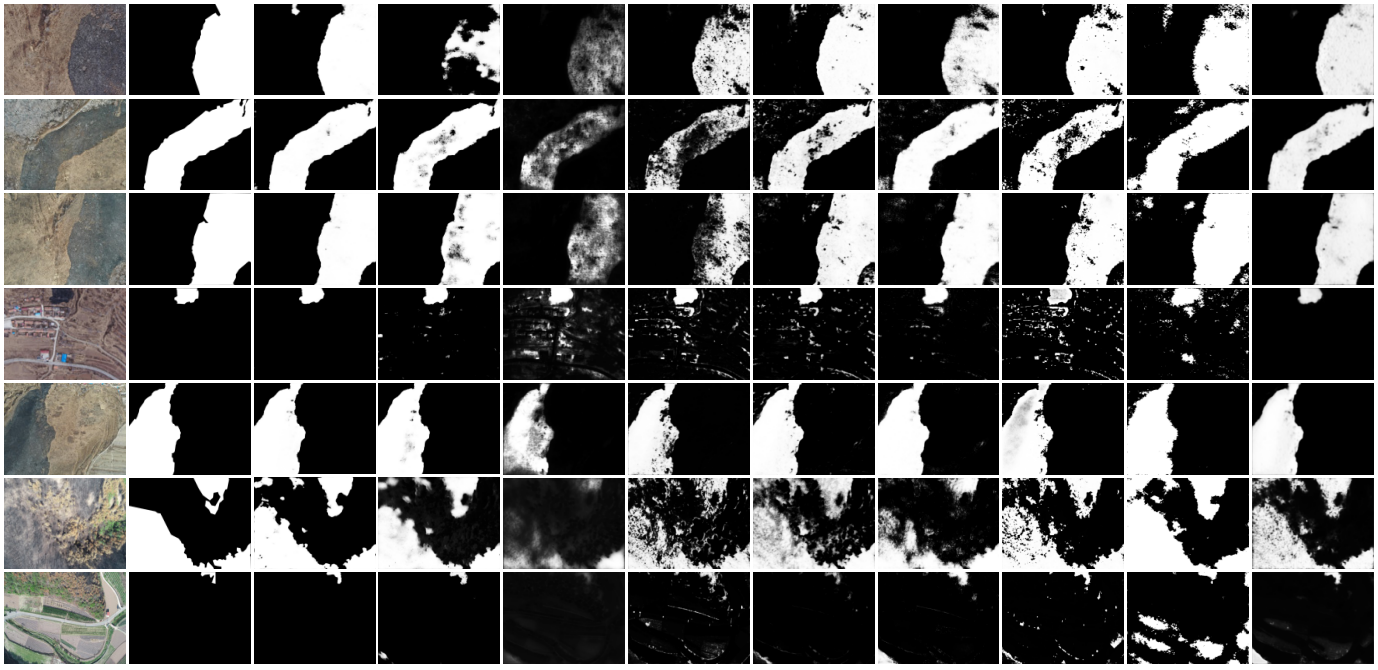


Fig. 9. Visual comparison of saliency maps. 1) Chongli District dataset (first five rows). 2) Andong City dataset (the last two rows). The columns are: (a) input images, (b) ground truth, (c) ours, (d) CTDNet, (e) CPD, (f) F3Net, (g) GateNet, (h) LDF, (i) MiNet, (j) GLSTR, and (k) GCPANet.

Saliency TRansformer (GLSTR) [42], where all results are generated from source code or provided by the creators of the SOD methods. The deep learning-based methods are trained using the same training data as the proposed methods with the default parameter settings of the corresponding models.

*1) Qualitative Comparison:* The results of different methods are shown in Fig. 9. The figure shows five challenging scenarios, including regions with complex edges, regions with similar backgrounds, tiny regions, large regions, and regions with more occlusions. Our method shows superiority and competitiveness in the following aspects.

*a) Advantages in positioning accuracy:* Our method is capable of accurately locating prominent targets with superior background interference suppression. In the fourth image (row 4 of Fig. 9), due to the local weather and geography of the scene, there are numerous detected areas that are composed of vegetation similar to the target area. The other deep learning-based networks more or less detect these indistinguishable regions. In contrast, our network suppresses all the background noise and identifies the target burned areas. In the seventh image (row 7), the target area is very small, while there are hard-to-identify background noise regions introduced by lighting during image capture. Regardless of whether the target area is very large or very small, our network accurately locates the burned region and segments it. Thus, our network has great advantages in terms of localization accuracy, and it has effectively suppressed background noise as well as precise localization ability.

*b) Advantages in content integrity:* In the first image (row 1 of Fig. 9), the burned area to be detected is very large. As well as being influenced by geographical factors, there are many scattered tiny regions on it that are different from the burned area, i.e., noise existing in the target area. From the comparison results, it can be seen that other segmentation networks have difficulty in segmenting the target area as a whole accurately, while our network segments the whole area as an entire significant target. The result of our network is more valuable as a reference for post-fire management and recovery. All prediction networks in the second, third, fifth, and sixth images basically suffer from the problem of detecting incomplete target regions. However, our proposed network ensures that the segmented significant target regions are more holistic and complete to the greatest extent.

*c) Advantages in boundary details:* In existing BAS methods, researchers usually addressed the problem that aerial images taken by UAV are too large to be segmented. However, since the BAS makes the image lose a lot of global semantic information and spatial location information, the resultant target region boundary is poorly segmented. Fig. 9 also shows that our network detected the edge detail well. A SOD method has the advantage of utilizing spatial location information, and compared with Fig. 10, it can be seen that the network with SOD has better edge detail. Thus, the advantage of our network in terms of edge detail is important.

*2) Quantitative Comparison:* The P-R curves in Fig. 11 show that our BASNet outperforms the other methods, achieving the highest accuracy on both datasets. Table I reports the quantitative performances of the different methods. It shows that the performance of our method is significantly better in all performance measures than the state-of-the-art SOD and semantic segmentation methods for BAS on the two datasets. In addition, Table III shows that BASNet has a greater

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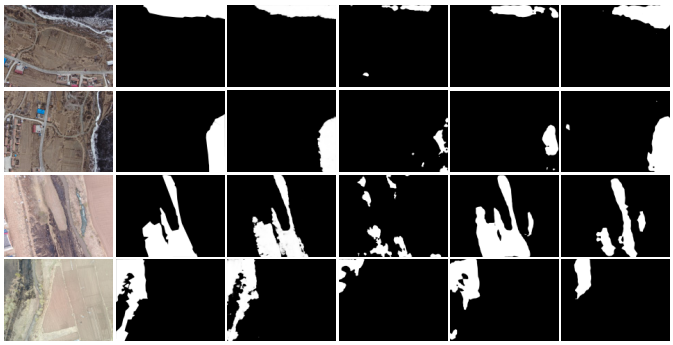
TABLE III
QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART SOD MODELS ON OUR TWO DATASETS IN TERMS OF PARAMETERS AND FPS

| Methods | F3Net | LDF | CTDNet | MINet | GateNet | CPD | GLSTR | GCPANet | BASNet(ours) |
|---|---|---|---|---|---|---|---|---|---|
| Parameters | 25.54 | 25.15 | 11.82 | 162.38 | 128.63 | 29.10 | 162.98 | 67.05 | 13.62 |
| FPS | 2 | 2 | 5 | 0.5 | 1 | 2 | 5 | 1 | 5 |



(a) (b) (c) (d) (e) (f)

Fig. 10. Visual comparison with several semantic segmentation networks. The columns are: (a) input images, (b) ground truth, (c) ours, (d) UNet, (e) PSPNet, and (f) DANet.

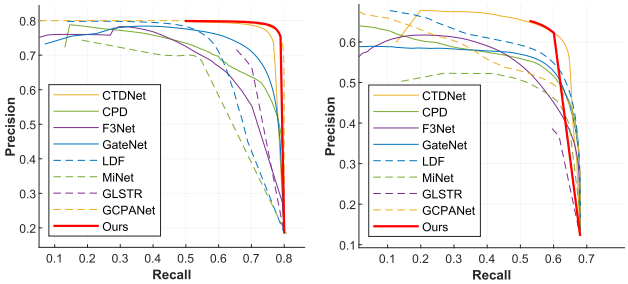
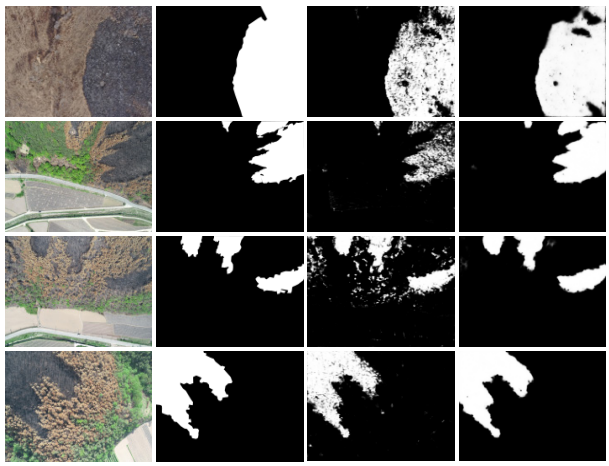



Fig. 11. P-R curves on the dataset of China and South Korea. (a) China. (b) South Korea.

advantage in prediction speed and number of parameters among the networks it compares with.

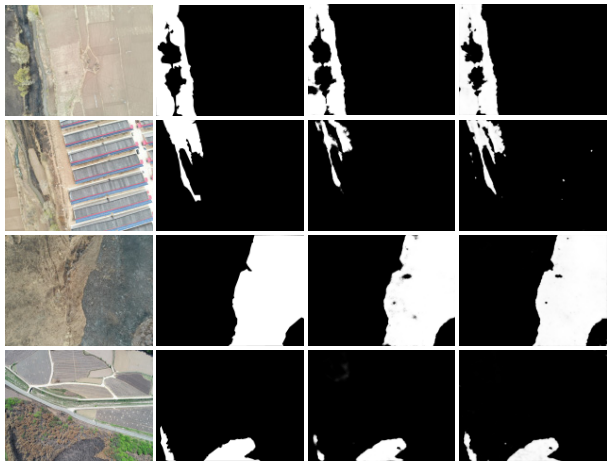
### D. Comparision With Semantic Segmentation Methods

We have also conducted experiments on some mainstream semantic segmentation networks including UNet [34], pyramid scene parsing network (PSPNet) [43], dual attention network (DANet) [44]. The quantitative results are shown in Table I, and the visualization of the results are shown in Fig. 10. It can be seen that semantic segmentation performs poorer for regions with complex boundaries and less severe burns such as the several scenarios illustrated in this article. Furthermore, semantic segmentation networks consume more memory during training. Based on the experimental observation, the input images with the resized size of $1024 \times 1024$ already occupies more than 90% memory when using 1080Ti GPU. This means the size of images must be reduced significantly, where some semantic information and spatial information might be lost. High memory consumption also introduces some difficulties in the actual deployment of the network.



(a) (b) (c) (d)

Fig. 12. Visual comparison of saliency maps in adding RFM in the ablation study. (a) Input images. (b) Ground truth. (c) Base. (d) Base+RFM.



(a) (b) (c) (d)

Fig. 13. Visual comparison of saliency maps in adding the CB in the ablation study. (a) Input images. (b) Ground truth. (c) Base+RFM+GAP. (d) Base+RFM+GAP+CB.

### E. Ablation Study

We validated the effectiveness of each module of BASNet on both two datasets by adding them to the baseline network gradually. We also verified the contribution of using API loss. The same training parameters are used in all experiments (see Section IV-B). The backbone is ResNet18 and is pretrained on ImageNet.

Fig. 12 shows that the proposed RFM effectively suppresses background noise, extracts more accurate features, and generates a more complete overall saliency map when compared with the baseline net. In addition, Fig. 13 shows that CB

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TABLE IV

ABLATION STUDIES FOR DIFFERENT MODULES: CHANGES IN THE THREE MAIN EVALUATION METRICS ON TWO DATASETS AS EACH MODULE IS GRADUALLY ADDED

| Module | | | | | | CHINA | | | SOUTH KOREA | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| base | RFM | GAP | CB | CBAM | BRM | $F_\beta$ | MAE | $E_m$ | $F_\beta$ | MAE | $E_m$ |
| ✓ | | | | | | 0.705 | 0.038 | 0.717 | 0.468 | 0.041 | 0.557 |
| ✓ | ✓ | | | | | 0.752 | 0.023 | 0.752 | 0.563 | 0.018 | 0.629 |
| ✓ | ✓ | ✓ | | | | 0.767 | 0.016 | 0.756 | 0.567 | 0.021 | 0.632 |
| ✓ | ✓ | ✓ | ✓ | | | 0.750 | 0.015 | 0.761 | 0.580 | 0.021 | 0.637 |
| ✓ | ✓ | ✓ | ✓ | ✓ | | 0.759 | 0.016 | 0.759 | 0.578 | 0.018 | 0.625 |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | **0.772** | **0.010** | **0.767** | **0.615** | **0.016** | **0.657** |

TABLE V

ABLATION STUDIES FOR DIFFERENT MODULES: CHANGES IN THE NUMBER OF PARAMETERS AND FPS ON THE TWO DATASETS

| Module | | | | | | | Parameters | FPS |
|---|---|---|---|---|---|---|---|---|
| base | RFM | GAP | CB | CBAM | BRM | 1X1 | | |
| ✓ | | | | | | | 12.56M | 13 |
| ✓ | ✓ | | | | | | 12.73M | 11 |
| ✓ | ✓ | ✓ | | | | | 12.73M | 11 |
| ✓ | ✓ | ✓ | ✓ | | | | 21.70M | 7 |
| ✓ | ✓ | ✓ | ✓ | ✓ | | | 21.71M | 5 |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | 21.71M | 4 |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 13.62M | 5 |

integrates richer semantic information and refines more details via RFM, which further improves the accuracy of the saliency map.

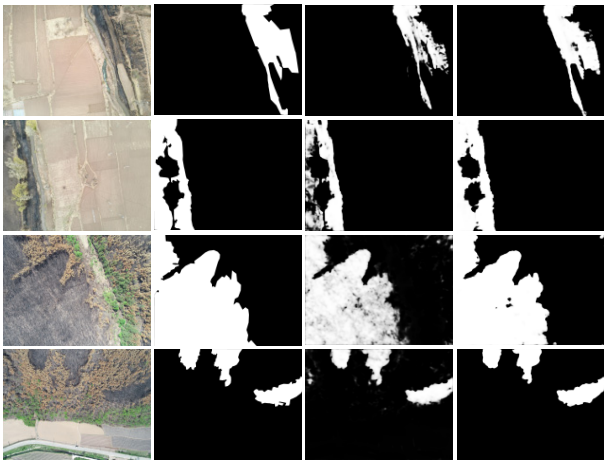
*1) Comparison of Accuracy:* Table IV shows that the feature fusion capability of RFM greatly improves the overall prediction accuracy of the network. This shows how the complementary fusion of features is crucial in the detection of salient targets for large-scale images. It is important to note that the components of the network have a complementary role with each other. For example, the performance of the network is improved less when CBAM is added directly but is greatly improved after cooperating with the new module behind the network. Therefore, it can be concluded that each component of the network not only plays a role in improving the network performance individually, but also performs better when they are integrated. This demonstrates the effectiveness of each component of our network and the overall effectiveness of the network.

*2) Comparison of Speed:* Table V shows that in the process of gradually adding the network modules, the speed of the network decreases from 13 to 4 frames per second (FPS) and the number of parameters increases from 12.56 to 21.71 M. The added $1 \times 1$ convolution reduces the parameters of the network to 13.62 M, the speed increases to 5 FPS, and all the performance metrics are improved. The $1 \times 1$ convolution effectively diminishes the redundant information of the network's multiple channels and improves its performance. Reducing the memory requirement of the device enables our proposed method to achieve better adaptability in different application scenarios.

TABLE VI

QUANTITIVE COMPARISON OF EFFECTS ON THE API IN THE LOSS FUNCTION

| Loss functions | CHINA | | | SOUTH KOREA | | |
|---|---|---|---|---|---|---|
| | $F_\beta$ | MAE | $E_m$ | $F_\beta$ | MAE | $E_m$ |
| BCE+IOU | 0.726 | 0.031 | 0.738 | 0.522 | 0.032 | 0.592 |
| API | 0.772 | 0.010 | 0.767 | 0.615 | 0.016 | 0.657 |



(a) (b) (c) (d)

Fig. 14. Visual comparison of saliency maps in adding the API loss. (a) Input images. (b) Ground truth. (c) BCE+IOU. (d) API.

*3) Comparison of With/Without Using API Loss:* We also conducted experiments to validate the use of API loss, and the results are shown in Table VI. The table shows that API loss improves the overall performance of BASNet significantly more than directly summing up the BCE and IoU losses. In addition, we also visualized the difference with or without using API loss in Fig. 14. The figure shows that the use of API loss refines the edges of the burned areas predicted by the network and enhances the overall integrity of the prediction results.

### *F. Failure Cases*

Fig. 15 shows some failure cases of our proposed method. There are more complex distractors in the burned areas in these scenes, or there are nonsignificant areas that are easily

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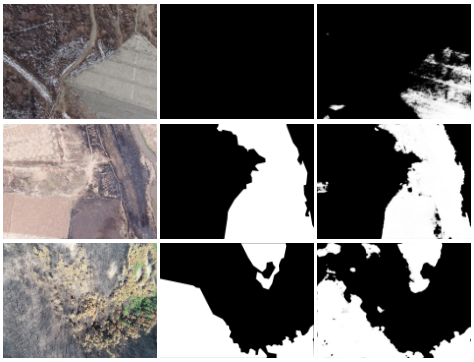


Fig. 15. Some failure cases of our proposed method. (a) Input images. (b) Ground truth. (c) BASNet.

misclassified by the network, such as in the first row of Fig. 15. Such scenes can cause the network to mis-detect false positive regions, resulting in unsatisfactory results. In addition, complex scenes can lead to reduced completeness of the segmentation results, which is challenging for our current network. In the future, we will further improve our network to achieve better results, starting from the detailed information of salient targets and local association information.

## V. Conclusion

In this article, the method of SOD is applied for the first time for BAS, and an efficient and accurate network is proposed for this scenario. The network is divided into two submodules. The positioning module efficiently extracts high-level semantic features and global contextual information via the GAP layer and the CB to determine the approximate location of salient regions. The network is further improved using RFM with API loss. The experimental results show that the application of BASNet to BAS effectively addressed the problems of region occlusions, similar background noise, blurred boundaries, slow operation speed, and the need for preprocessing or post-processing faced by existing methods. The results have also been significantly improved. This approach also achieves the state-of-the-art performance on two of the latest UAV-acquired burned areas datasets.

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